

Exponential Functions

$$\checkmark \frac{d}{dx}(e^x) = e^x$$

$$\checkmark \frac{d}{dx}(a^x) = a^x \ln a$$

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} f'(x)$$

$$\frac{d}{dx}(a^{f(x)}) = \ln(a) a^{f(x)} f'(x)$$

Logarithmic Functions

$$\checkmark \frac{d}{dx}(\ln x) = \frac{1}{x}, x > 0$$

$$\frac{d}{dx} \ln(g(x)) = \frac{g'(x)}{g(x)}$$

$$\checkmark \frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}, x > 0$$

$$\frac{d}{dx}(\log_a g(x)) = \frac{g'(x)}{g(x) \ln a}$$

Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

Inverse Trigonometric Functions

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, x \neq \pm 1$$

$$\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, x \neq \pm 1$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}, x \neq \pm 1, 0$$

$$\frac{d}{dx}(\csc^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}, x \neq \pm 1, 0$$

Hyperbolic Functions

$$\frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\cosh x) = \sinh x$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$$

$$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

$$\frac{d}{dx}(\coth x) = -\operatorname{csch} x$$

Inverse Hyperbolic Functions

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

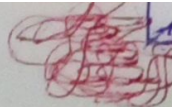
$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}, x > 1$$

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}, |x| < 1$$

$$\frac{d}{dx}(\operatorname{csch}^{-1} x) = \frac{-1}{|x|\sqrt{1-x^2}}, x \neq 0$$

$$\frac{d}{dx}(\operatorname{sech}^{-1} x) = \frac{-1}{x\sqrt{1-x^2}}, 0 < x < 1$$

$$\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}, |x| > 1$$


$$\int \cos x \, dx = \sin x + c$$

$$\int 1 \, dx = x + c$$

$$\int \sin x \, dx = -\cos x + c$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + c$$

$$\int \tan x \, dx = \ln |\sec x| + c$$

$$\int \cot x \, dx = \ln |\sin x| + c$$

$$\int \operatorname{cosec} x \, dx = \ln |\operatorname{cosec} x - \cot x| + c$$

$$\rightarrow \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\rightarrow \lim_{x \rightarrow 0} (1+ax)^{1/x} = e^a$$

$$\rightarrow 1 - \cos 2\theta = 2\sin^2\theta$$

$$\rightarrow \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$

$$\rightarrow \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\rightarrow \Gamma n = \int_0^{\infty} e^{-x} x^{n-1} dx \quad n > 0$$

$$\rightarrow \beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad m, n > 0$$

$$\rightarrow \beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$$

$$\rightarrow \Gamma n+1 = n \Gamma n \quad \rightarrow \int_0^{\infty} x^{1/4} e^{-\sqrt{x}} dx$$

$$\rightarrow \Gamma n+1 = n!$$

$$\rightarrow \frac{\beta(m+1, n)}{\beta(m, n)} = \frac{m}{m+n}$$

$$\rightarrow \Gamma 1 = 1$$

$$\rightarrow \Gamma 1/2 = \sqrt{\pi}$$

$$\Gamma n = (n-1)\Gamma n-1 \quad | \quad \Gamma n-1 = (n-2)\Gamma n-2$$

$$\Gamma n-3 = (n-4)\Gamma n-4$$

$$\Rightarrow K = \frac{dy}{ds}$$

K = curvature, s = arc length

$$\Rightarrow \rho = \frac{1}{K}$$

ρ = radius of curvature

$$\Rightarrow K = \frac{\frac{d^2y}{dx^2}}{\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{3/2}}$$

$$\Rightarrow K = \frac{y''}{(1 + y'^2)^{3/2}}$$

$$\Rightarrow \rho = \frac{1}{K}$$

$$\Rightarrow \rho = \frac{(1 + y'^2)^{3/2}}{y''}$$

$$\Rightarrow K = \frac{r^2 + 2\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2r}{d\theta^2}}{\left\{r^2 + \left(\frac{dr}{d\theta}\right)^2\right\}^{3/2}} \quad \text{OR} \quad K = \frac{r^2 + 2r'^2 - r r''}{(r^2 + r'^2)^{3/2}}$$

$$\Rightarrow \rho = \frac{1}{K}$$

$$\text{OR } \rho = \frac{(r^2 + r'^2)^{3/2}}{r^2 + 2r'^2 - r r''}$$

$$\Rightarrow 1 + \cot^2 x = \operatorname{Cosec}^2 x$$

$$\Rightarrow \int_0^{\pi/2} \sin^p x \cos^q x dx = \frac{\left|\frac{p+1}{2}\right| \left|\frac{q+1}{2}\right|}{2 \sqrt{\frac{p+q+2}{2}}}$$

$$\rightarrow \int \sin^n x \, dx = \frac{-\cos x \sin^{n-1} x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$\rightarrow \int_0^{\pi/2} \sin^n x \, dx = \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{2}{3} ; n \text{ is odd} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{1}{2} \cdot \frac{\pi}{2} ; n \text{ is even} \end{cases}$$

$$\rightarrow \int \cos^n x \, dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$$\rightarrow \int_0^{\pi/2} \cos^n x \, dx = \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{2}{3} ; n \text{ is odd} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{1}{2} \cdot \frac{\pi}{2} ; n \text{ is even} \end{cases}$$

$$\rightarrow \int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx$$

$$\rightarrow \int \cot^n x \, dx = -\frac{\cot^{n-1} x}{n-1} - \int \cot^{n-2} x \, dx$$

$$\rightarrow \int \sec^n x \, dx = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$$

$$\rightarrow \int \operatorname{cosec}^n x \, dx = \frac{-\cot x \operatorname{cosec}^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \operatorname{cosec}^{n-2} x \, dx$$

$$\rightarrow \int \sin^p x \cdot \cos^q x \, dx = -\frac{\cos^{q+1} x \sin^{p-1} x}{p+q} + \frac{p-1}{p+q}$$

$$\int \sin^{p-2} x \cos^q x \, dx$$

$$\rightarrow (x+y)^n = x^n + n x^{n-1} y + \frac{n(n-1)}{2!} x^{n-2} y^2 + \dots + y^n$$

$${}^n C_0 x^n y^0 + {}^n C_1 x^{n-1} y^1 + {}^n C_2 x^{n-2} y^2 + \dots + {}^n C_n x^0 y^n$$

$$\rightarrow x^n + \frac{1}{x^n} = 2 \cos n\theta$$

$$\rightarrow x^n - \frac{1}{x^n} = 2i \sin n\theta$$

$$\rightarrow x^n = \cos n\theta + i \sin n\theta = (\cos \theta + i \sin \theta)^n$$

+ve $\rightarrow$ -ve max

$$\begin{aligned} \int x^n dx &= \frac{x^{n+1}}{n+1} + c \quad \text{or} \quad \int (ax+b)^n dx = \frac{1}{a} \frac{(ax+b)^{n+1}}{n+1} + c \\ \int \frac{1}{x} dx &= \ln x + c \quad \text{or} \quad \int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + c \\ \int \sin x dx &= -\cos x + c \\ \int \cos x dx &= \sin x + c \\ \int \sec^2 x dx &= \tan x + c \\ \int \operatorname{cosec}^2 x dx &= -\cot x + c \\ \int \tan x \sec x dx &= \sec x + c \\ \int \operatorname{cosec} x \cot x dx &= -\operatorname{cosec} x + c \\ \int \cot x dx &= \ln(\sin x) + c \\ \int \sec x dx &= \ln|\sec x + \tan x| + c \\ \int \operatorname{cosec} x dx &= \ln|\operatorname{cosec} x - \cot x| + c \\ \int \tan x dx &= \ln|\sec x| + c \\ &\quad \text{or} \quad -\ln|\cos x| + c \end{aligned}$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{x}{a}$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{a^2 + x^2})$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a}$$

$$\int f'(x) f^{-1}(x) dx = \ln f(x) + c$$

Definite Integrals

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_a^b f(x) = \int_a^c f(x) + \int_c^b f(x) dx$$

Trigonometric Substitutions

$$\sqrt{a^2 - x^2} \quad x = a \sin \theta$$

$$\sqrt{a^2 + x^2} \quad x = a \tan \theta$$

$$\sqrt{x^2 - a^2} \quad x = a \sec \theta$$

Integration by Parts

$$\int u dv = uv - \int v du$$

$$\int a^x dx = \frac{1}{\ln a} a^x + c$$

$$\int \sin \theta dx = -\cos \theta + c$$

$$\int a^x [f(x) + g(x)] dx = \int a^x f(x) dx + \int a^x g(x) dx$$

surface Area of rectangular box = $2(lw + lh + wh)$

Gamma Function:

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx, n > 0$$

Beta Function $\beta(m, n)$

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

m, n are +ve, m, n > 0

Relation b/w α & β

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\Gamma(1) = 1$$

$$\beta(m, m) = \int_0^1 \frac{x^{m-1}}{(1+x)^{m+1}} dx$$

$$\Gamma(n+1) = n \Gamma(n)$$

$$\Gamma(n+1) = n!$$

$$\Gamma(n) = (n-1) \Gamma(n-1)$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\int_0^{\pi/2} \sin^n x = \frac{\sqrt{\frac{n+1}{2}} \sqrt{\pi}}{\sqrt{\frac{n+2}{2}}}$$

$$\int_0^{\pi/2} \cos^n x dx = \frac{\sqrt{\frac{n+1}{2}} \sqrt{\pi}}{2 \sqrt{\frac{n+2}{2}}}$$

$$\sec^2 z = 1 + \tan^2 z \quad \sin(-z) = -\sin z$$

$$\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$$

$$\tan(z_1 + z_2) = \frac{\tan z_1 + \tan z_2}{1 - \tan z_1 \tan z_2}$$

$$\sin z = \sin \bar{z}$$

$$\cos z = \cos \bar{z}$$

$$\int_0^{\pi/2} \sin^p x \cos^q x = \frac{\sqrt{\frac{p+1}{2}} \sqrt{\frac{q+1}{2}}}{2 \sqrt{\frac{p+q+2}{2}}}$$

λ = modulus, $|z| = \sqrt{a^2 + b^2}$
 θ = argument = $\tan^{-1}\left(\frac{b}{a}\right)$
 $= \text{Arg}(z)$

range of complex number = principle argument, $-\pi < \theta < \pi$

$$a = \lambda \cos \theta, b = \lambda \sin \theta$$

$$Z = x + yi$$

$$\lambda \cos \theta + i \lambda \sin \theta$$

$$\lambda (\cos \theta + i \sin \theta)$$

$$\lambda (e^{i\theta})$$

DE MOUVRE'S THEOREM

$$(x^n = \cos n\theta + i \sin n\theta)$$

$$\rightarrow x^n = (\cos \theta + i \sin \theta)^n$$

$$\rightarrow \frac{1}{x^n} = \cos n\theta - i \sin n\theta$$

$$\rightarrow x^n + \frac{1}{x^n} = 2 \cos n\theta$$

$$\rightarrow x^n - \frac{1}{x^n} = 2i \sin n\theta$$

EXPONENTIAL FUNCTION

$$e^z = e^{x+yi} = e^x \cdot e^{yi}$$

$$= e^x (\cos y + i \sin y)$$

$$e^{iy} = \cos y + i \sin y$$

$$e^{-iy} = \cos y - i \sin y$$

$$e^{iy} = \cos y + i \sin y$$

$$\cos y = \frac{e^{iy} + e^{-iy}}{2}$$

$$\sin y = \frac{e^{iy} - e^{-iy}}{2i}$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\tan z = \frac{e^{iz} - e^{-iz}}{i(e^{iz} + e^{-iz})}$$

HYPERBOLIC FUNCTIONS

$$\sinh z = \frac{e^z - e^{-z}}{2}$$

$$\tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\cosh z = \frac{e^z + e^{-z}}{2}$$

$$\cosh^2 z - \sinh^2 z = 1$$

$$\text{sech}^2 z = 1 - \tanh^2 z$$

$$\text{cosech}^2 z = \coth^2 z - 1$$

$$\sinh(z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2$$

$$\tanh(z_1 + z_2) = \frac{\tanh z_1 + \tanh z_2}{1 + \tanh z_1 \tanh z_2}$$

FROM CIRCULAR TO HYPERBOLIC

$$\sin(iz) = i \sinh z$$

$$\cos(iz) = \cosh z$$

$$\tan(iz) = i \tanh z$$

$$\cot(iz) = -i \coth z$$

$$\text{cosec}(iz) = -i \text{cosech} z$$

$$\sec(iz) = \text{sech} z$$

FROM HYPERBOLIC TO CIRCULAR

$$\sinh(iz) = i \sin z$$

$$\cosh(iz) = \cos z$$

$$\tanh(iz) = i \tan z$$

$$\coth(iz) = -i \cot z$$

$$\text{cosech}(iz) = -i \text{cosec} z$$

$$\text{sech}(iz) = \sec z$$